

# Health facts: Gas-powered data centres - CAPE

Category: Environmental, Water Table, Heat & Wildlife Impacts | Evidence

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## Source Content

Every day doctors see how damage to the environment harms human health. When industries pollute the air, water, and soil, our patients pay the price. That's why the Canadian Association of Physicians for the Environment in Alberta wants to share what we know about gas-powered data centres. AI data centres need a lot of energy to run, so Alberta's current electrical grid cannot support them. Instead, they will be fuelled by on-site natural gas generators. This type of generator has been studied extensively, so we know there are health risks for communities that live near them. Read more about these risks below or download our fact sheet. Communities can protect themselves by demanding evidence-based health impact assessments before approving new data centres. Communities deserve to know how proposed data centres might damage the air, water and soil that they depend on and a real plan to protect their health.

**Air Pollution** - Natural gas generators release several types of air pollution, including:

- Nitrous oxides, which have been linked to life-threatening breathing problems like asthma and chronic bronchitis and chronic cognitive disorders like dementia.
- Airborne particulate matter, which are particles small enough to enter not only the lungs but the blood stream and contribute to serious cardiovascular conditions like heart attacks and strokes.
- Volatile organic compounds, which are airborne chemicals like benzene, formaldehyde, and acetaldehyde. Breathing in these compounds is known to increase the risk of several types of cancer.
- This combination of pollution has been linked to earlier deaths, and long-term exposure can cause harm even when emissions are below regulatory limits.
- Because of their high energy needs, large AI data centres often have back up diesel generators. Diesel is known to cause cancer and emits other harmful pollutants like carbon monoxide, which can cause fatal poisonings at high doses.

**Forever Chemicals** - Data centres also use PFAS, or "forever chemicals," in their cooling systems and electronic components, which can leak into the air, soil and water.

- When communities are exposed, these long-lasting chemicals have been linked to cancer, birth defects, thyroid disease, and weakened immune systems.

**Water Use** - Data centres use tremendous amounts of energy, and need to be actively cooled. This cooling uses millions of litres of water per day.

- Communities in Alberta that are already impacted by drought could experience increased strain on water infrastructure with multiple health impacts, including stress related to poor agricultural outputs and food insecurity.

**Noise Pollution** - Data centres that operate continuously are large-scale industrial operations that generate significant environmental noise.

- Long-term exposure to noise pollution has been linked to poor sleep, headaches, high blood pressure, heart disease, mental distress, learning difficulties, and decreased quality of life.

**Overburdened Healthcare Infrastructure** - Most data centres are planned for rural areas where large areas of land are available for their development.

Rural communities in Alberta already face significant challenges with access to healthcare, including family doctor shortages and emergency room closures. - Large scale infrastructure projects like data centres will bring in new workers and new health challenges. Without significant preparation, this could put more strain on already over-burdened health services. Vulnerable Populations - All of these health challenges will be most likely to impact vulnerable populations like young children, the elderly, pregnant people, and community members living with chronic illnesses or disabilities. Prepared by the Canadian Association of Physicians for the Environment Alberta Regional Committee

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